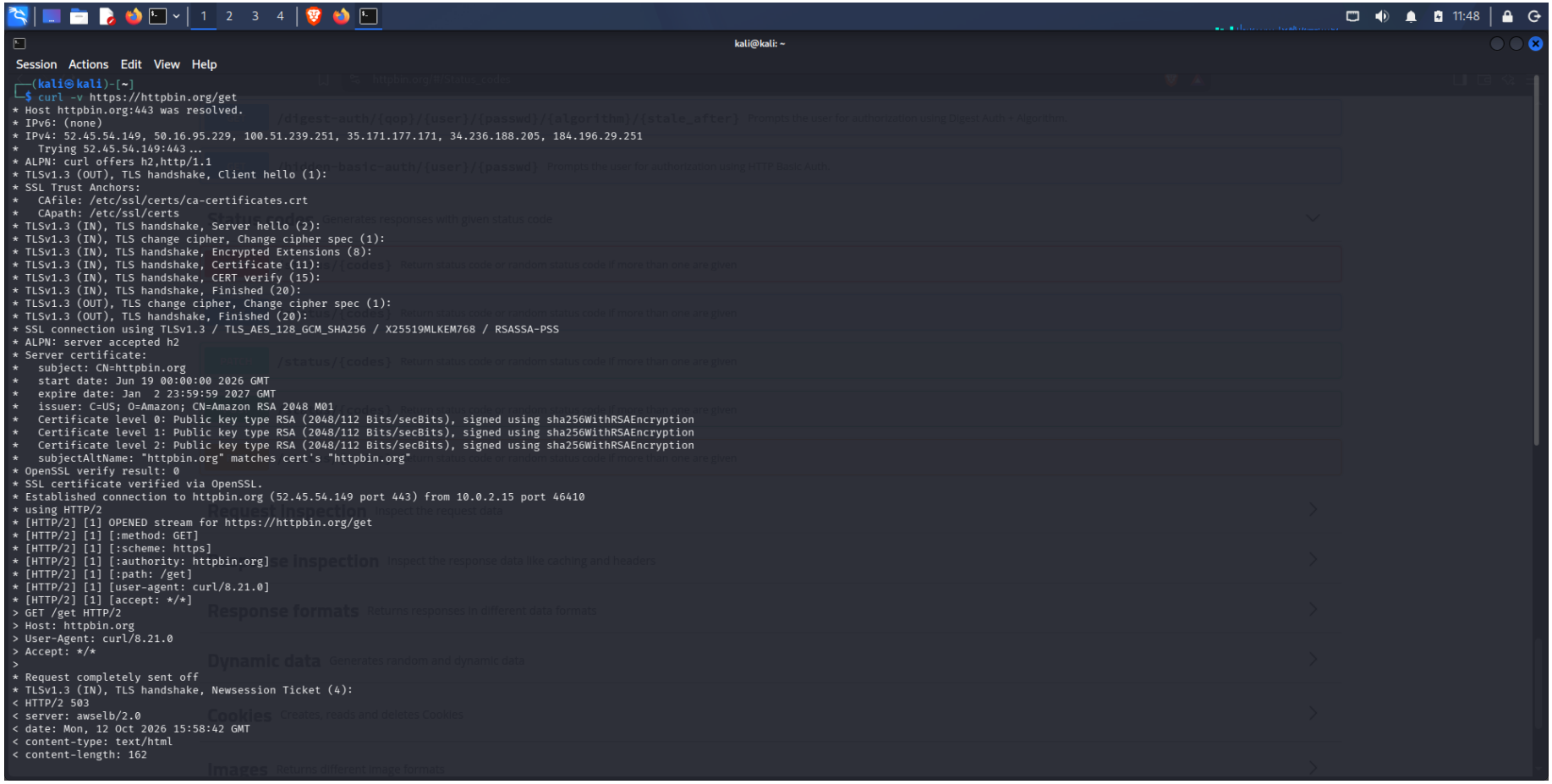
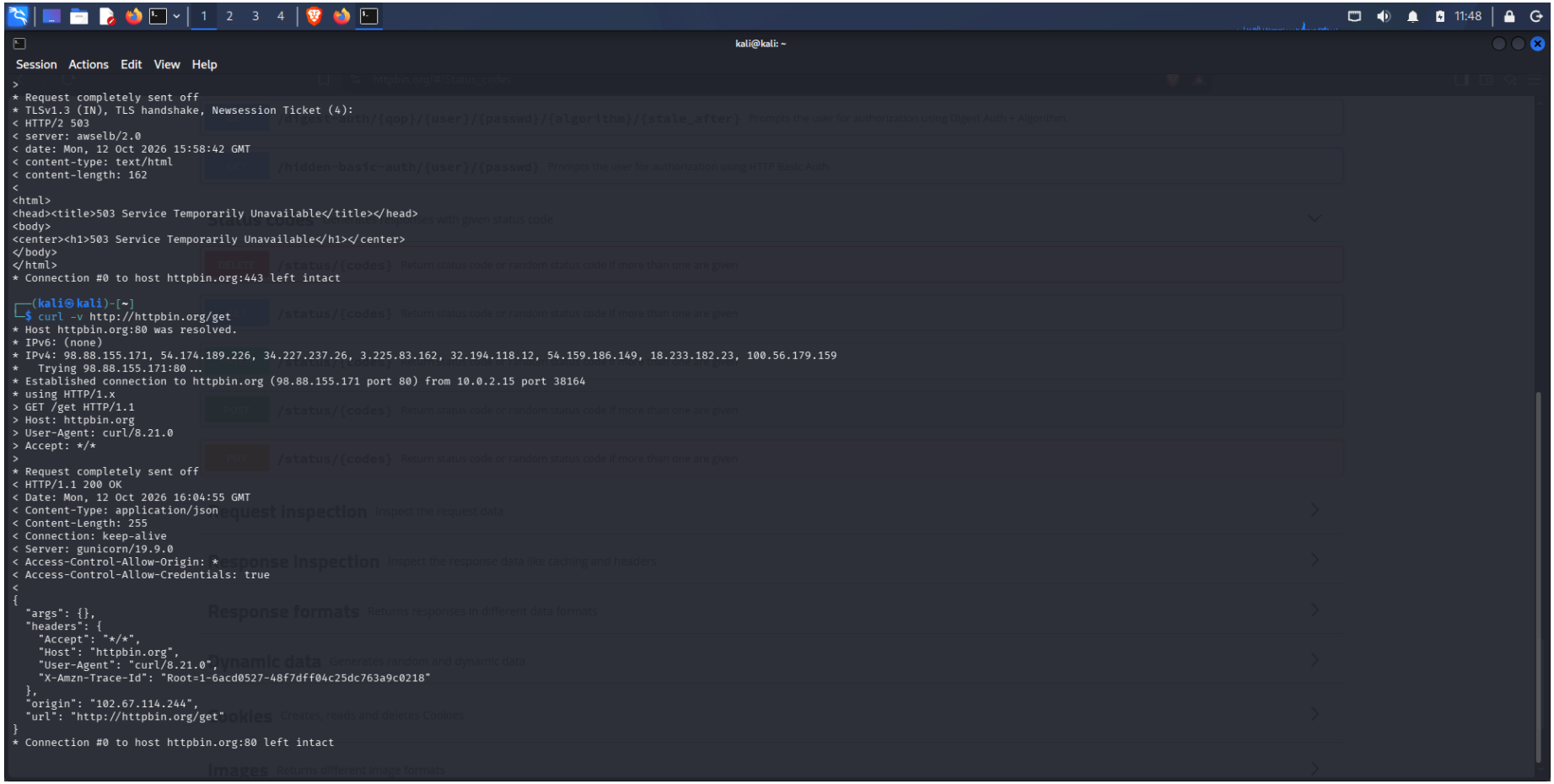

1 2 3 4
11:48
kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ curl -v https://httpbin.org/get
* Host httpbin.org:443 was resolved.
* IPv6: (none)
* IPv4: 52.45.54.149, 50.16.95.229, 100.51.239.251, 35.171.177.171, 34.236.188.205, 184.196.29.251
*   Trying 52.45.54.149:443 ...
* ALPN: curl offers h2,http/1.1
* TLSv1.3 (OUT), TLS handshake, Client hello (1):
* SSL Trust Anchors:
*   CAfile: /etc/ssl/certs/ca-certificates.crt
*   CApath: /etc/ssl/certs
* TLSv1.3 (IN), TLS handshake, Server hello (2):
* TLSv1.3 (IN), TLS change cipher, Change cipher spec (1):
* TLSv1.3 (IN), TLS handshake, Encrypted Extensions (8):
* TLSv1.3 (IN), TLS handshake, Certificate (11):
* TLSv1.3 (IN), TLS handshake, CERT verify (15):
* TLSv1.3 (IN), TLS handshake, Finished (20):
* TLSv1.3 (OUT), TLS change cipher, Change cipher spec (1):
* TLSv1.3 (OUT), TLS handshake, Finished (20):
* SSL connection using TLSv1.3 / TLS_AES_128_GCM_SHA256 / X25519MLKEM768 / RSASSA-PSS
* ALPN: server accepted h2
* Server certificate:
*   subject: CN=httpbin.org
*   start date: Jun 19 00:00:00 2026 GMT
*   expire date: Jan 2 23:59:59 2027 GMT
*   issuer: C=US; O=Amazon; CN=Amazon RSA 2048 M01
*   Certificate level 0: Public key type RSA (2048/112 Bits/secBits), signed using sha256WithRSAEncryption
*   Certificate level 1: Public key type RSA (2048/112 Bits/secBits), signed using sha256WithRSAEncryption
*   Certificate level 2: Public key type RSA (2048/112 Bits/secBits), signed using sha256WithRSAEncryption
*   subjectAltName: "httpbin.org" matches cert's "httpbin.org"
* OpenSSL verify result: 0
* SSL certificate verified via OpenSSL.
* Established connection to httpbin.org (52.45.54.149 port 443) from 10.0.2.15 port 46410
* using HTTP/2
* [HTTP/2] [1] OPENED stream for https://httpbin.org/get
* [HTTP/2] [1] [:method: GET]
* [HTTP/2] [1] [:scheme: https]
* [HTTP/2] [1] [:authority: httpbin.org]
* [HTTP/2] [1] [:path: /get]
* [HTTP/2] [1] [user-agent: curl/8.21.0]
* [HTTP/2] [1] [accept: */*]
> GET /get HTTP/2
> Host: httpbin.org
> User-Agent: curl/8.21.0
> Accept: */*
>
* Request completely sent off
* TLSv1.3 (IN), TLS handshake, Newsession Ticket (4):
< HTTP/2 503
< server: awselb/2.0
< date: Mon, 12 Oct 2026 15:58:42 GMT
< content-type: text/html
< content-length: 162


kali@kali: ~
Session Actions Edit View Help
>
* Request completely sent off
* TLSv1.3 (IN), TLS handshake, Newsession Ticket (4):
< HTTP/2 503
< server: awselb/2.0
< date: Mon, 12 Oct 2026 15:58:42 GMT
< content-type: text/html
< content-length: 162
<
<html>
<head><title>503 Service Temporarily Unavailable</title></head>
<body>
<h1>503 Service Temporarily Unavailable</h1>
</body>
</html>
* Connection #0 to host httpbin.org:443 left intact
┌──(kali㉿kali)-[~]
└─$ curl -v http://httpbin.org/get
* Host httpbin.org:80 was resolved.
* IPv6: (none)
* IPv4: 98.88.155.171, 54.174.189.226, 34.227.237.26, 3.225.83.162, 32.194.118.12, 54.159.186.149, 18.233.182.23, 100.56.179.159
*   Trying 98.88.155.171:80 ...
* Established connection to httpbin.org (98.88.155.171 port 80) from 10.0.2.15 port 38164
* using HTTP/1.x
> GET /get HTTP/1.1
> Host: httpbin.org
> User-Agent: curl/8.21.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Mon, 12 Oct 2026 16:04:55 GMT
< Content-Type: application/json
< Content-Length: 255
< Connection: keep-alive
< Server: gunicorn/19.9.0
< Access-Control-Allow-Origin: *
< Access-Control-Allow-Credentials: true
<
{
  "args": {},
  "headers": {
    "Accept": "*/*",
    "Host": "httpbin.org",
    "User-Agent": "curl/8.21.0",
    "X-Amzn-Trace-Id": "Root=1-6acd0527-48f7dff04c25dc763a9c0218"
  },
  "origin": "102.67.114.244",
  "url": "http://httpbin.org/get"
}
* Connection #0 to host httpbin.org:80 left intact

TLS (Transport Layer Security) is the security protocol that protects communication between a web browser and a web server in HTTPS.

TLS plays three main roles:

1. Confidentiality (Encryption): Encrypts data exchanged between the browser and server to prevent unauthorized parties from reading it.
2. Authentication: Uses digital certificates to help the browser verify the server's identity and confirm it is communicating with the intended website.
3. Integrity: Detects unauthorized modification of data while it is transmitted.

In summary: HTTPS = HTTP + TLS. TLS protects HTTP communications against eavesdropping, impersonation, and tampering while data travels across the network.